Programming Engineering

Course 11 – 12 May

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  - Creational Patterns
  - Structural Patterns
  - Behavioral Patterns
  - Other Design Patterns
- Quality Assurance
  - Software Testing
  - Testing Methodologies
  - Testing process
  - Manual Testing vs. Automatic Testing

Recap

- GOF = ?
- Creational Patterns
- Structural Patterns
- Behavioral Patterns

Recap – CP

- Abstract Factory
- Builder
- Factory Method
- Prototype
- Singleton

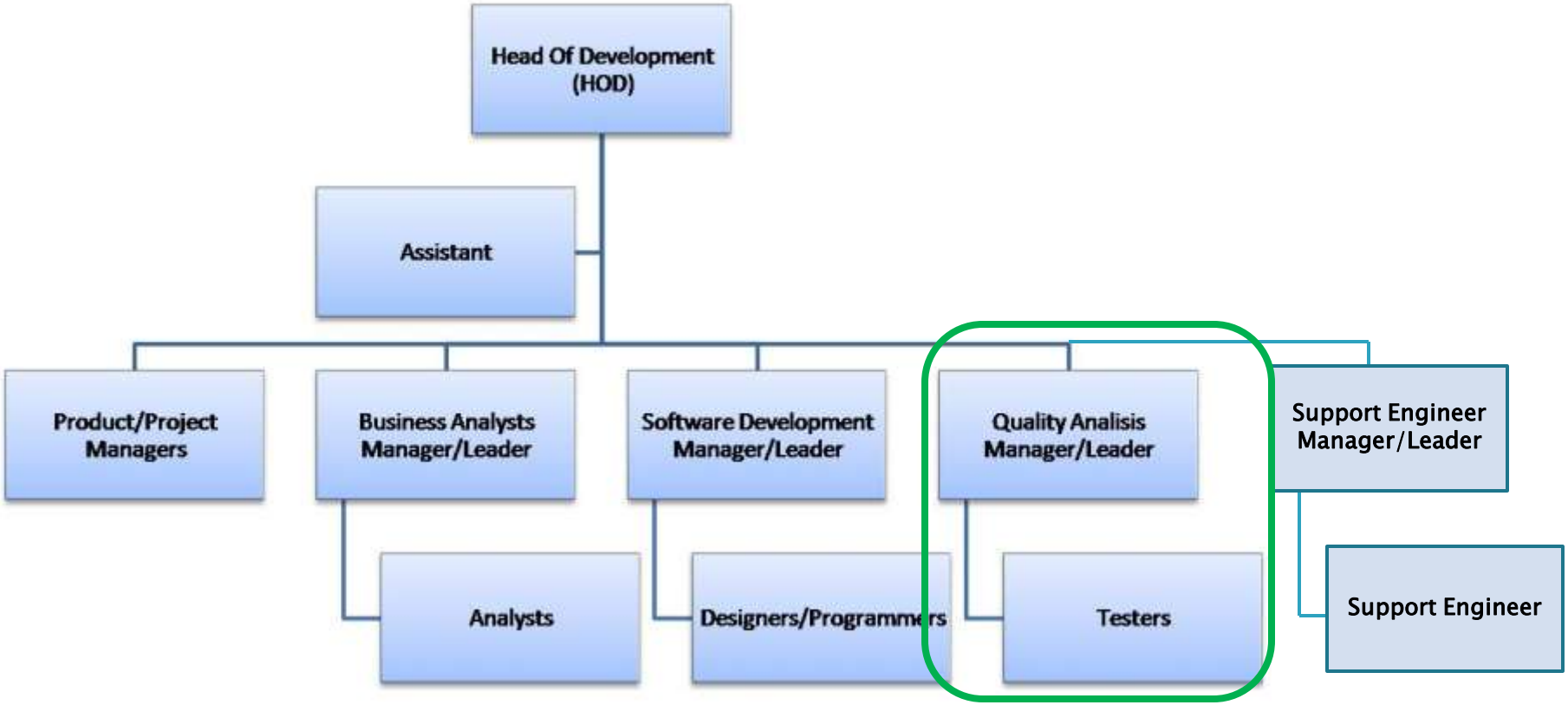
Recap – SP

- Adapter
- Bridge
- Composite
- Decorator
- Façade
- Flyweight
- Proxy

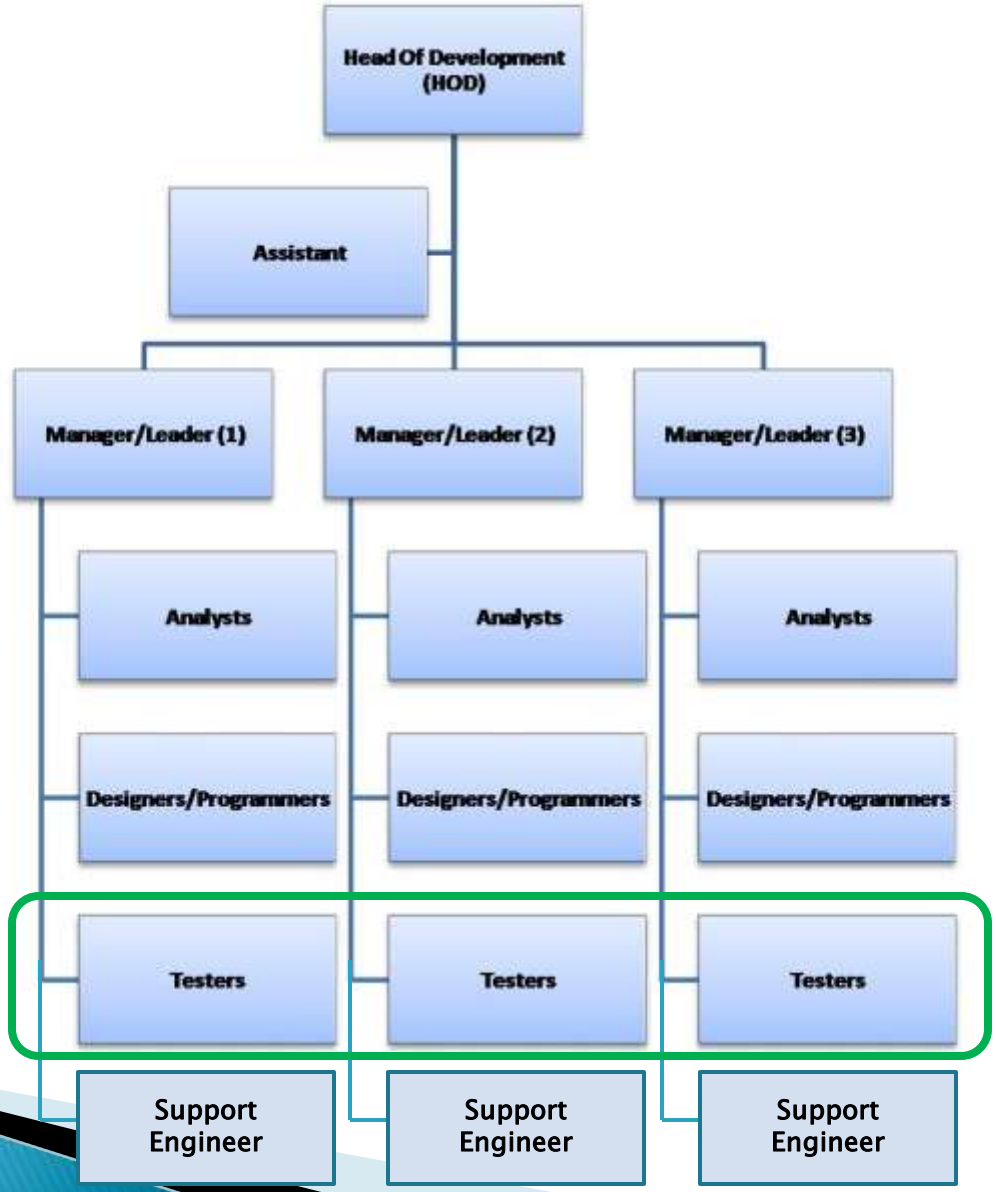
Recap – BP

- Chain of Responsibility
- Command
- Interpreter
- Iterator
- Mediator
- Memento
- Observer
- State
- Strategy
- Template Method
- Visitor

Hierarchy of an IT Company – Compartments



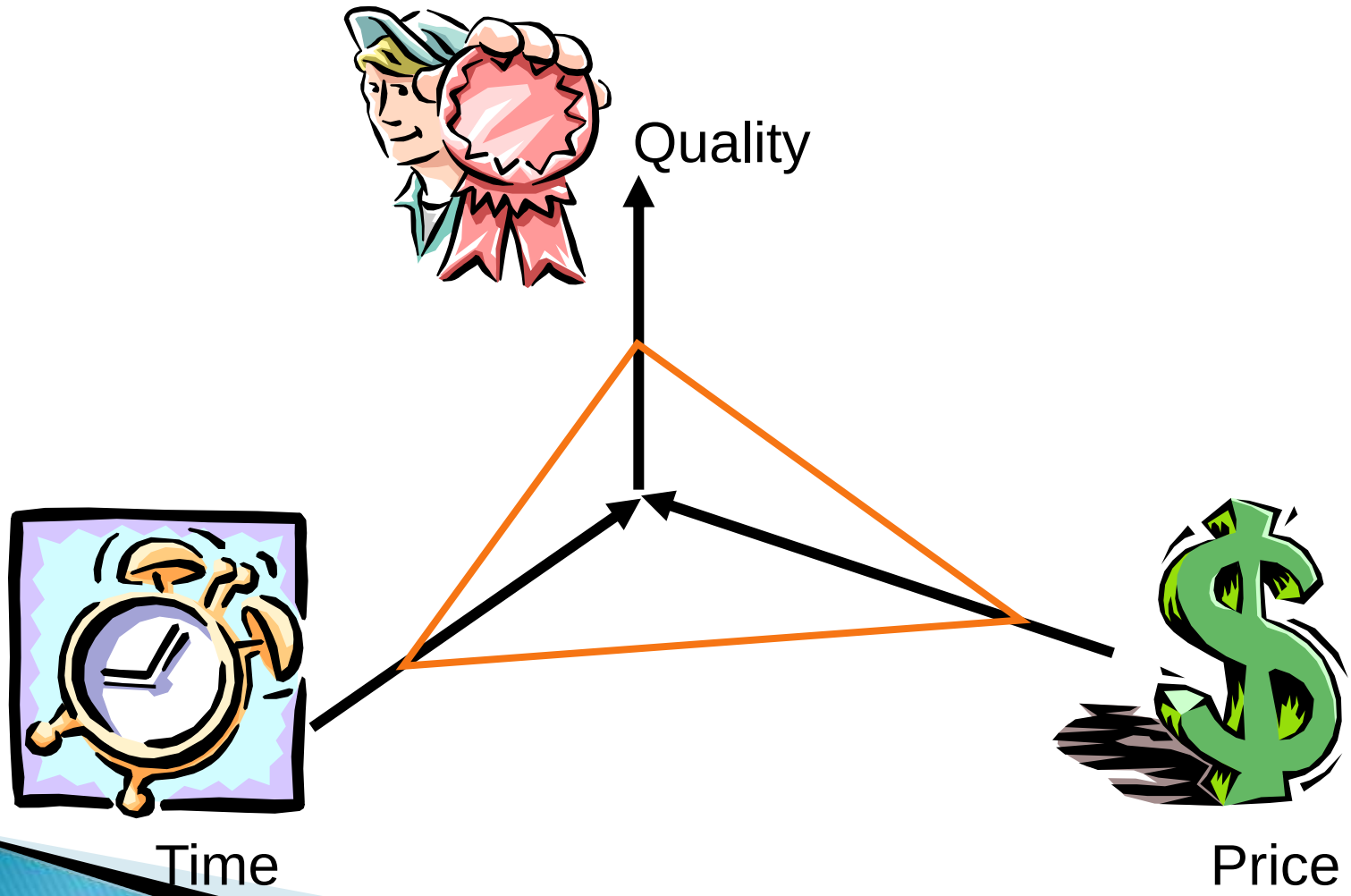
Hierarchy of an IT Company – Projects



Quality Assurance

- **Planned and systematic production processes** that provide confidence in a product's suitability for its intended purpose.
- A set of activities intended to ensure that **products satisfy customer requirements**
- *QA cannot absolutely guarantee the production of quality products but makes this more likely*
- Two key principles of QA:
  - "fit for purpose" – the product should be suitable for the intended purpose, and
  - "right first time" – mistakes should be eliminated

Quality Assurance Dilemma



Quality Assurance – Definition

"The process of exercising or evaluating a system by manual or automated means to verify that it satisfies specified <u>requirements</u> or to identify differences between expected and actual results."

(IEEE Standard Glossary, 1983)

Software Quality Assurance

- (SQA) consists of a means of **monitoring the software engineering processes and methods used to ensure quality**
- May include ensuring conformance to one or more standards, such as ISO 9000 or CMMI

Software Quality Assurance

- SQA encompasses the entire software development process, which includes processes such as
  - *software design,*
  - *coding,*
  - *source code control,*
  - *code reviews,*
  - *testing,*
  - *change management,*
  - *configuration management, and*
  - *release management*

ISO 9000

- ISO 9000 is a family of standards for quality management systems
- Some of the requirements in ISO 9001 (from ISO 9000 family) include
  - a set of procedures;
  - monitoring processes;
  - keeping adequate records;
  - checking output for defects;
  - regularly reviewing individual processes;
  - facilitating continual improvement

Software Quality Assurance

- Quality assurance is concerned with prevention
  - Defect Prevention
  - Processes
  - Continuous improvement of this processes
- Quality control is concerned with correcting
  - Software testing is a subset of Quality control

Image credit test-institute.org

Software Testing – Introduction

- An empirical investigation conducted to provide information about the *quality of the product or service under test, with respect to the context in which it is intended to operate*.
- Allow the business to *appreciate and understand* the risks at implementation of the software
- Test techniques include the process of executing a program or application with **the intent of finding software bugs**
- The process of **validating and verifying** that a software program/application/product meets the business and technical requirements that guided its design and development

Software Testing – When?

- Can be implemented at any time in the development process
- However the most test effort is employed after the requirements have been defined and coding process has been completed

Software Testing

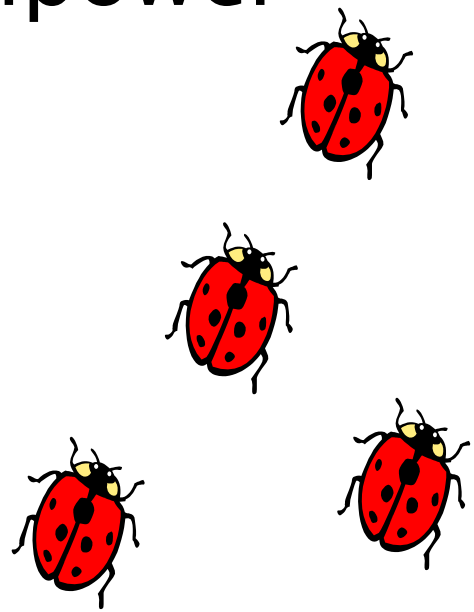
- Software Testing is NOT a phase
- It is integrated in all phases of software development
- Each development step has an attached testing documentation

What is the purpose of testing?

- We need not only to find bugs but also to prevent them (which is better)
- To know when to stop because effectiveness and economics of the process is essential.
- To know that not all system requires the same level of quality (mission critical against IT).
- Testing is not only for the SOFTWARE it is for all DELIVERABLES

Why are there bugs in software?

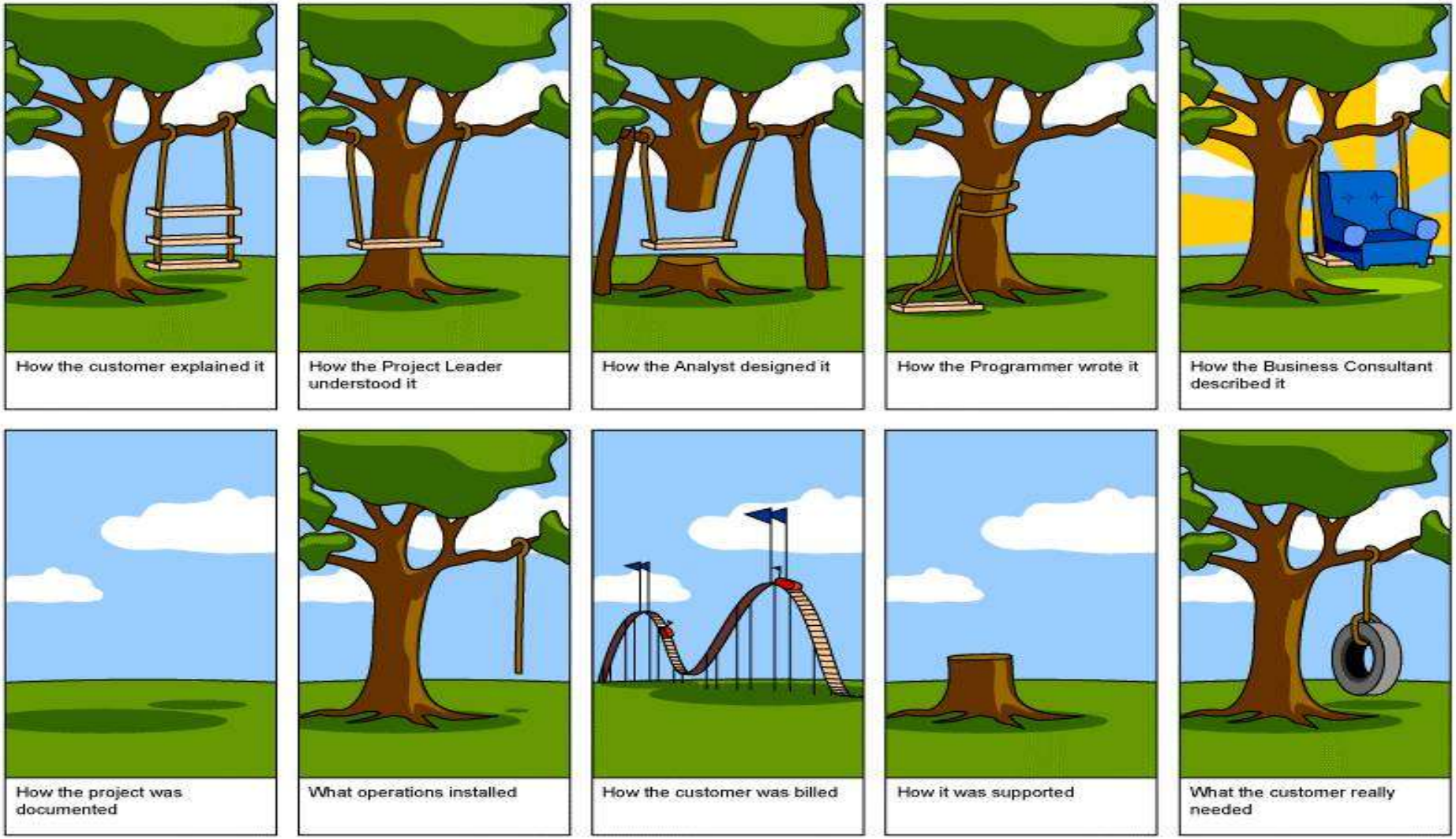
- Miscommunication
- Misunderstanding
- Low professional manpower
- Time pressures



Miscommunication

"Didn't you get my e-mail?"

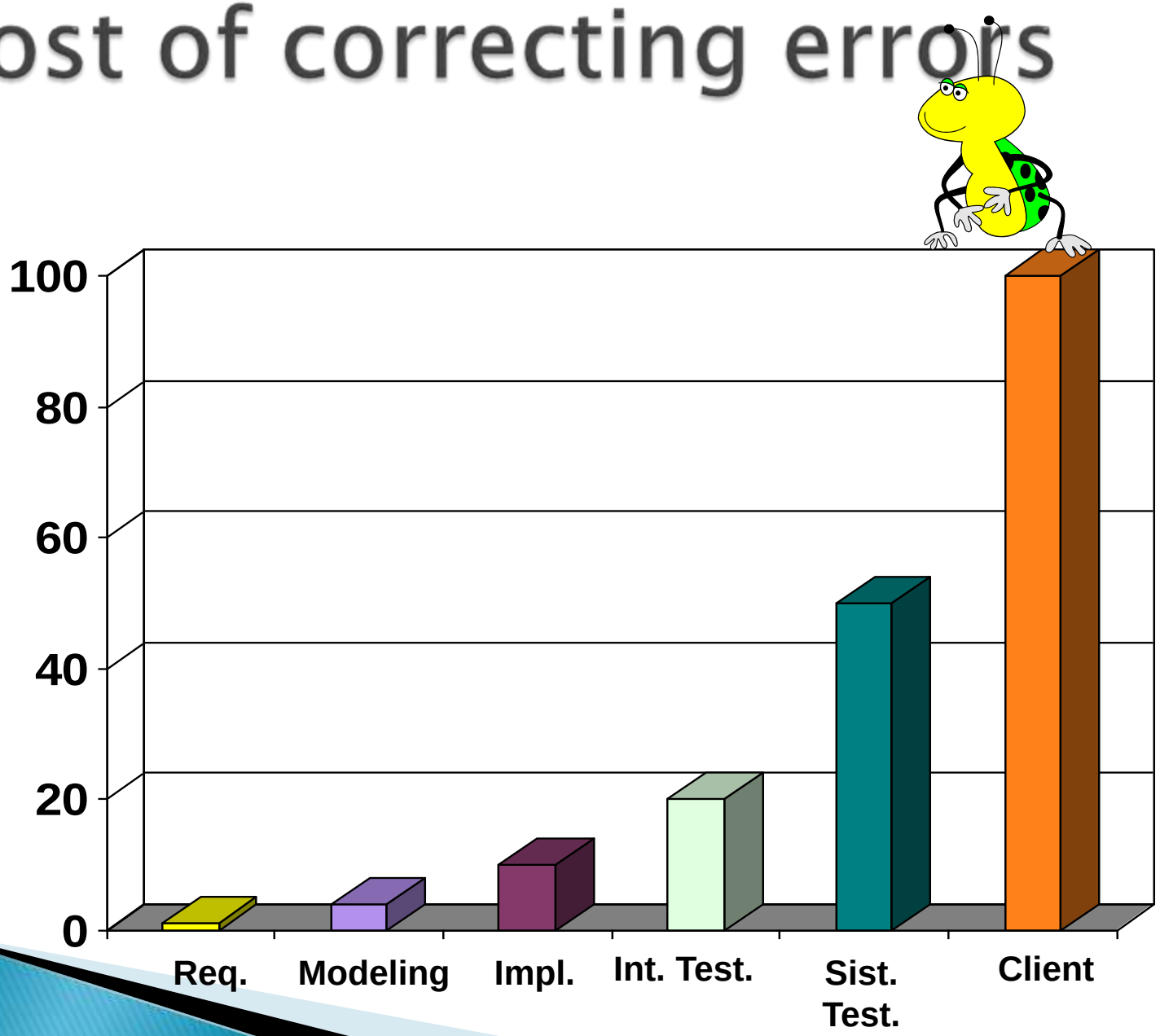
Misunderstandings



What genenerates most errors?

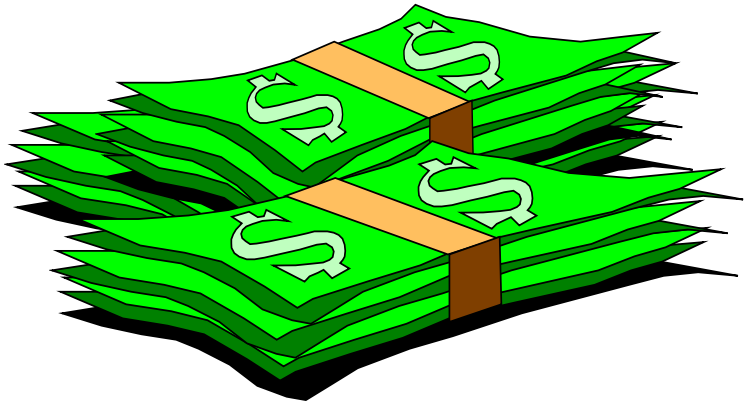
- Incorrect or incomplete requirements 50%
- Ambiguous or incomplete modeling 30%
- Programming errors 20%

Cost of correcting errors



Note

Late error detection⇒ greater repair cost



Error must be repaired as soon as possible

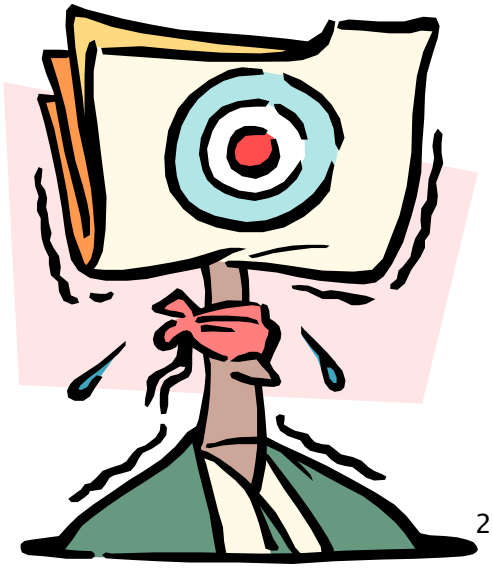
REQ. **MODELING** **IMPLEM.** **CLIENT TESTING**

Professional Testing

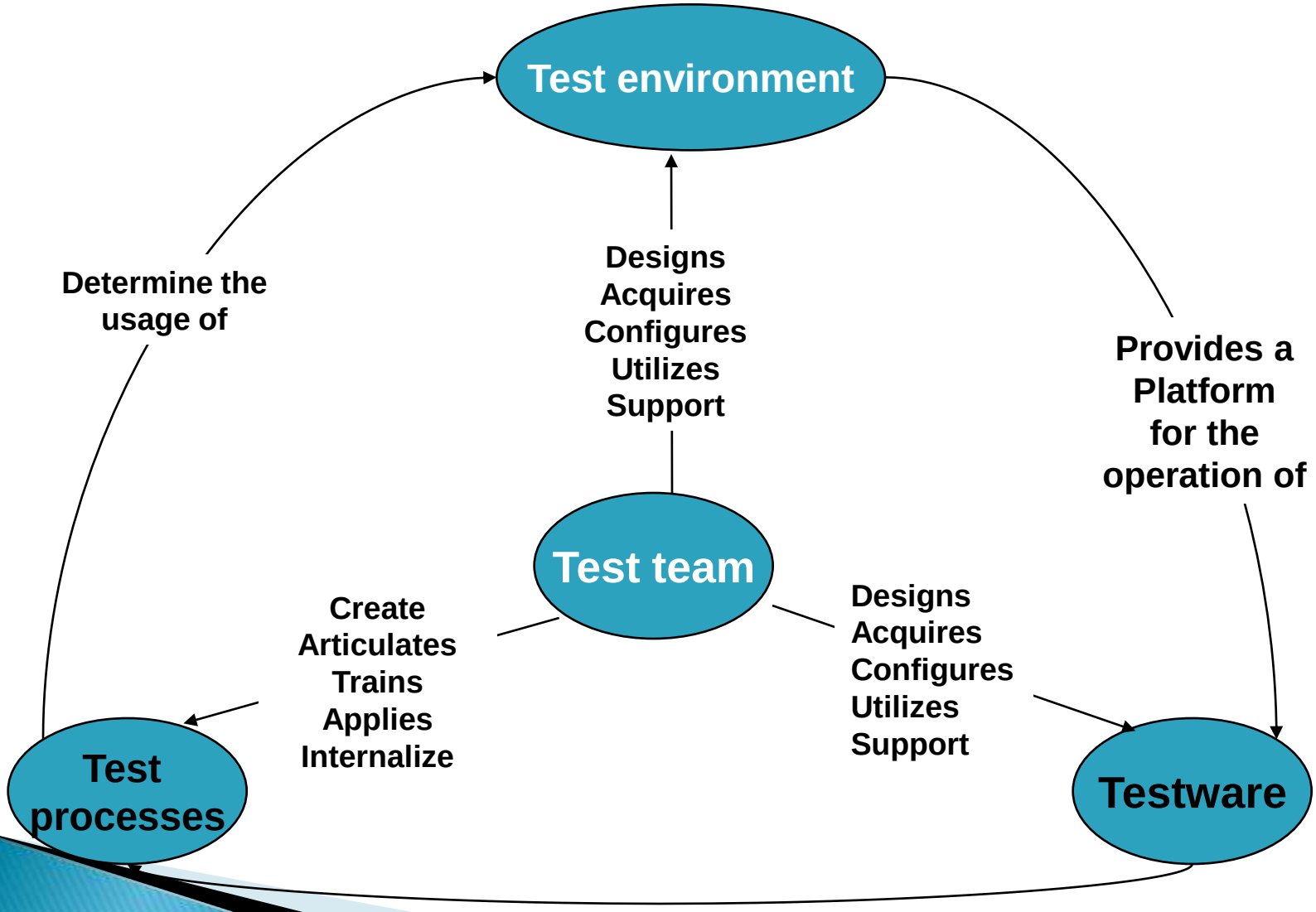
Professional testing means finding the least amount of test cases which will check the most amount of system features.

When does testing stop?

- Never
- When the number of errors found in a test cycle is lower than a set amount
- When no more critical faults are found
- When we run out of time



Schema of a Testing System



Testing Methodology

- Differences between testing and debugging
- Layers of testing
- Testing methods
- Testing content
- Manual vs Automatic Testing

Testing vs. Debugging

Testing	Debugging
• Check compliance to requirements	• Check validity of program sections
• Normally carried out by an external and neutral party	• Run by the developer
• Is planned and controlled	• Is a random process

Layers of testing

- Unit testing or debugging
- Module/Sub-System
- Integration
- System
- Acceptance

Unit Testing

- Testing a function, program screen, feature
- Run by programmers
- Predefined
- Results must be documented
- Input and Output simulators are used

Integration testing

- Testing of several modules at the same time
- Testing coexistence
- Run by programmers or testers
- Pre-planned testing
- Results must be documented

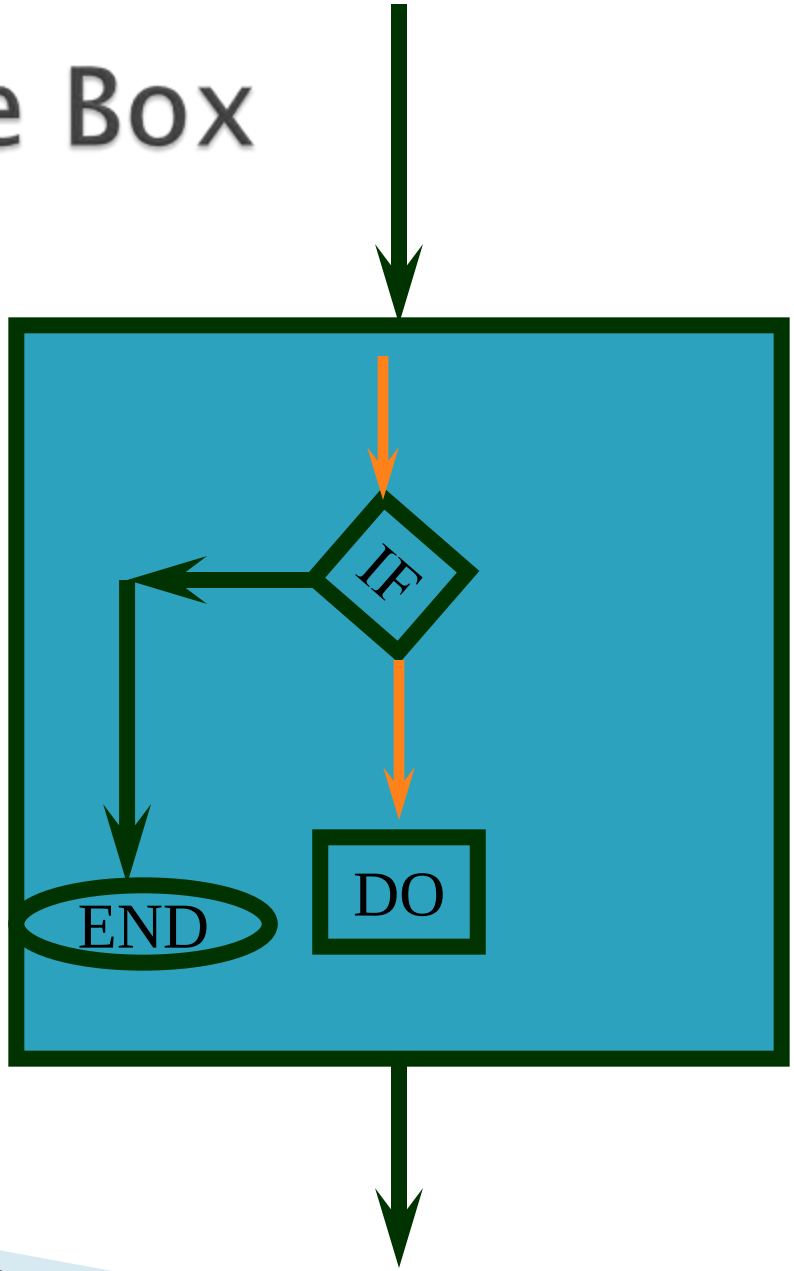
System Testing

- System testing of software or hardware is testing conducted **on a complete, integrated system** to evaluate the system's compliance with its specified requirements.
- System testing falls within the scope of **black box testing**
- System testing is a more limiting type of testing; it seeks to detect defects both within the "inter-assemblages" and also within the system as a whole.

Testing methods

- White Box
- Black Box
- Gray Box
- Graphical user Interface Testing
- Acceptance Testing
- Regression Testing

White Box

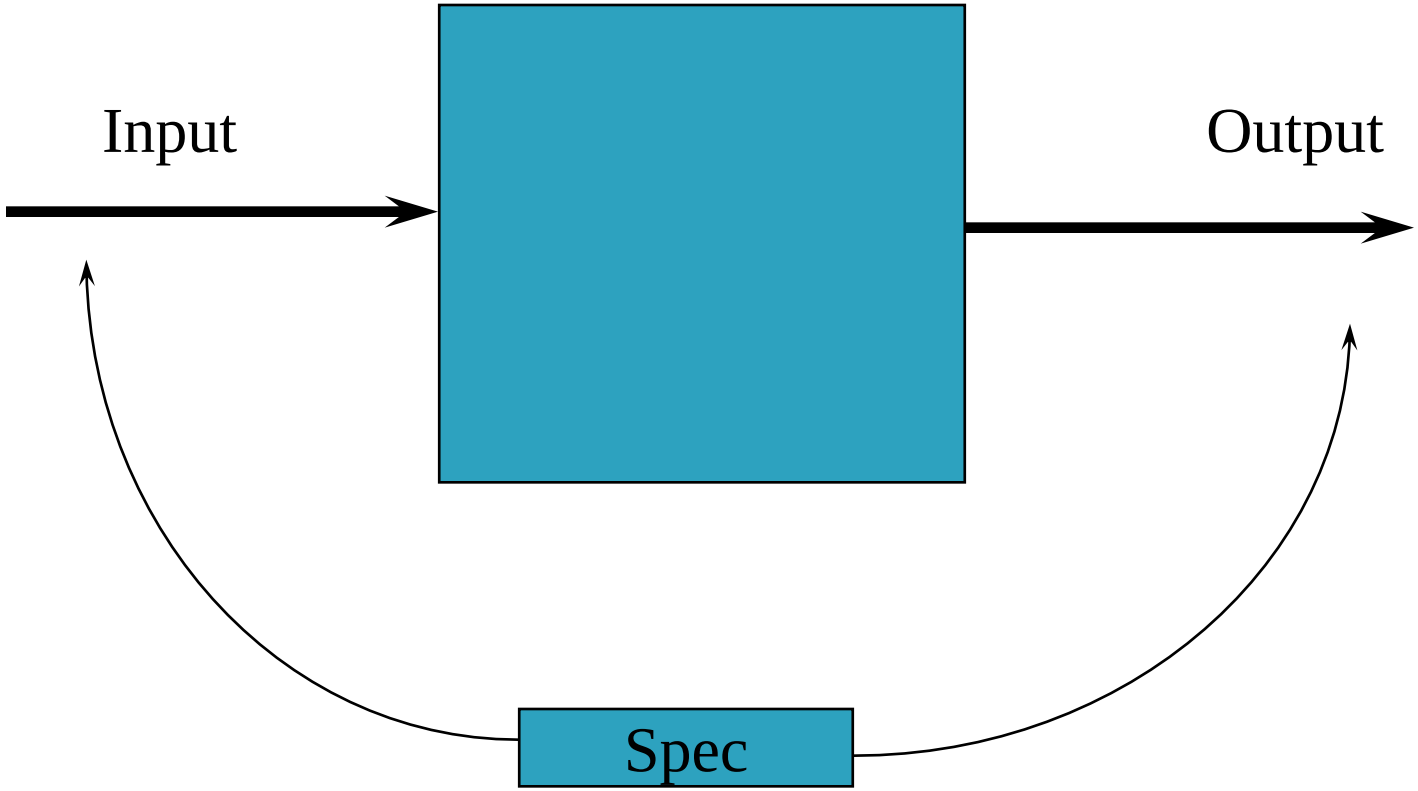


White Box (2)

- The tester has access to the internal data structures and algorithms

- Types of white box testing
  - api testing – Testing of the application using Public and Private APIs
  - code coverage – creating tests to satisfy some criteria of code coverage
  - fault injection methods
  - mutation testing methods
  - static testing – White box testing includes all static testing

Black Box



- "like a walk in a dark labyrinth without a flashlight,"

Black Box (2)

- Specification-based testing
- Black box testing methods include: equivalence partitioning, boundary value analysis, all-pairs testing, fuzzy testing, model-based testing, traceability matrix, exploratory testing and specification-based testing.

Grey Box

- This involves having access to internal data structures and algorithms for purposes of designing the test cases, but testing at the user, or black-box level
- Manipulating input data and formatting output do not qualify as "grey-box," because the input and output are clearly outside of the "black-box" that we are calling "the software under test"

GUI Testing

- In computer science, GUI software testing is the process of testing a product that uses a graphical user interface, to ensure it meets its written specifications.
- The variety of errors found in GUI applications:
  - Data validation, Incorrect field defaults, Mandatory fields, not mandatory, Wrong fields retrieved by queries, Incorrect search criteria
  - Field order, Multiple database rows returned, single row expected
  - Currency of data on screens, Correct window modality?
  - Control state alignment with state of data in window?

Acceptance Testing

- **A black-box testing performed on a system** prior to its delivery
- In software development, acceptance testing by the system provider is often distinguished from acceptance testing by the customer (the user or client) prior to accepting transfer of ownership.
- In such environments, acceptance testing performed by the customer is known as **user acceptance testing** (UAT).
- This is also known as **end-user testing**, site (acceptance) testing, or field (acceptance) testing.

Regression Testing

- Regression testing is **any type of software testing** which seeks to uncover software regressions.
- Such regressions occur whenever software functionality that was previously working correctly, stops working as intended.
- Typically regressions **occur as an unintended consequence of program changes**.
- Common methods of regression testing include **re-running previously run tests and checking whether previously fixed faults have re-emerged**

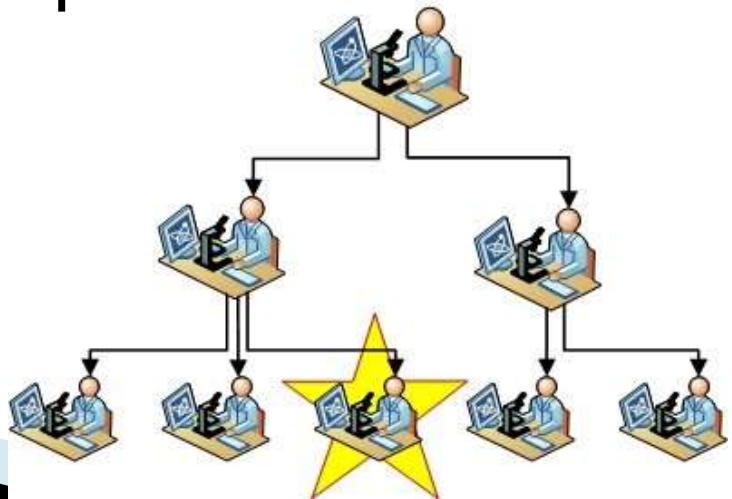
Automatic vs Manual Testing

- *Problems are found quickly*
- *Cheap to repeat*
- *The process of writing code is more flexible*
- *Less manual testing*
- *Software development becomes predictable and can be planned*

- *Solves interface issues: correctness of text, messages, page layout, element order, visibility etc.*
- *Writing test scenarios can be difficult and implies technical knowledge of the entire system*

Manual Testing

- *Manual testing is the process of manually testing software for defects*
- It requires a **tester** to play the role of an end user, and use most of all features of the application to ensure correct behavior
- To ensure completeness of testing, the tester often follows a written **test plan** that leads them through a set of important **test cases**



Definitions

- **Test Strategy** is developed by the "Project manager" which contains what type of technique to follow and which module to test
- **Test Plan** is developed by the Test Lead, which contains "what to test", "how to test", "when to test", "who to test"
- **Test Scenario** is a name given to test case. It is dealt with by the Test Engineer
- **Test Case** specifies a testable condition to validate functionality. The test cases are deal by Test Engineer

Test Strategy vs. Test Plan

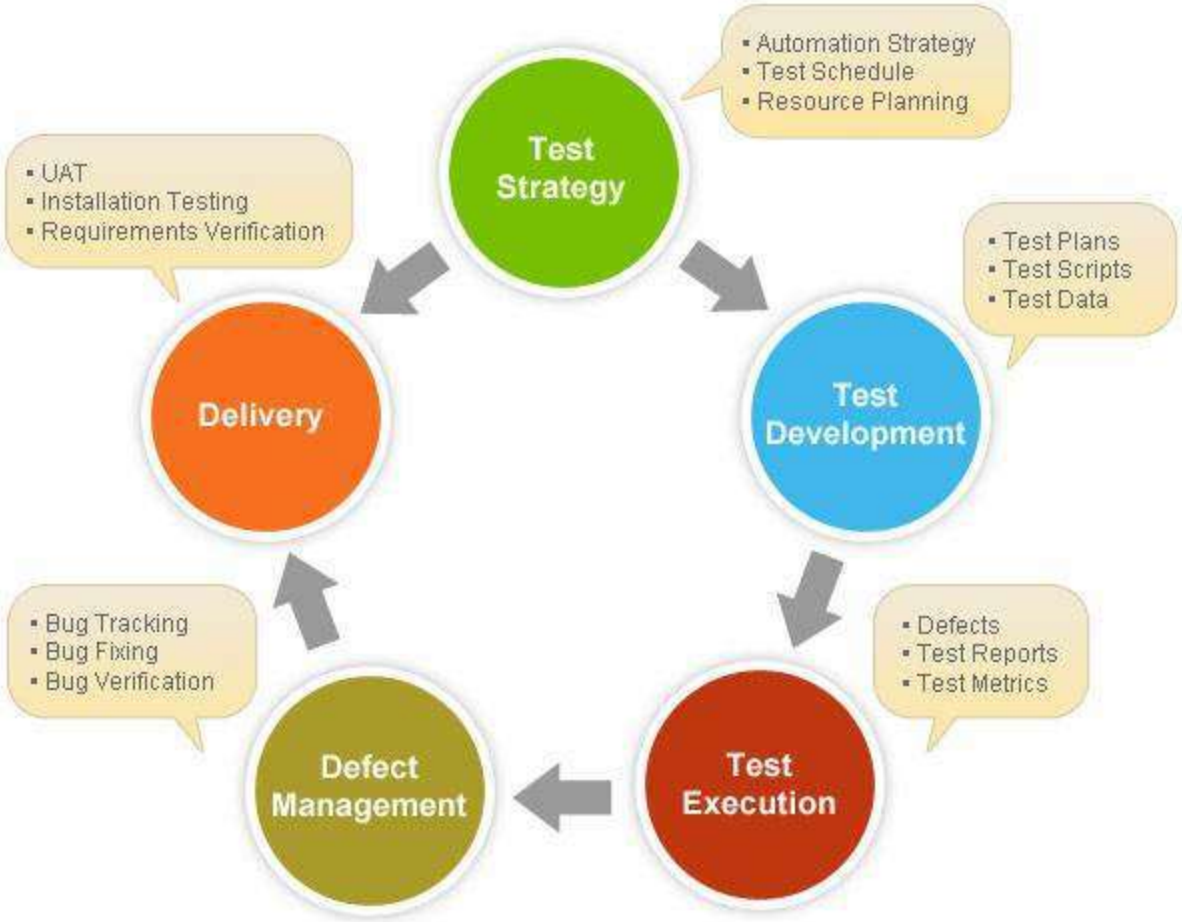


Image credit iconma.com

Test Plan

- **A systematic approach to testing a system**
- Contains a detailed understanding of what the eventual workflow will be
- Documents the strategy that will be used to verify and ensure that a product or system meets its design specifications and other requirements
- Is usually prepared by or with significant input from Test Engineers

Test Plan Components

- May include one or more of the following:
  - *Design Verification or Compliance test*
  - *Manufacturing or Production test*
  - *Acceptance or Commissioning test*
  - *Service and Repair test*
  - *Regression test*

Test Plan Structure (IEEE 829-1998)

- Test plan identifier
- Introduction
- Test items
- Features to be tested
- Features not to be tested
- Approach
- Item pass/fail criteria
- Suspension criteria
- Test deliverables
- Testing tasks
- Environmental needs
- Responsibilities
- Staffing and training needs
- Schedule
- Risks and contingencies
- Approvals

Test Plan – Life Cycle

Test Case

- **A set of conditions or variables** under which a tester will determine whether an application or software system meets specifications
- **A sequence of steps** to test the correct behavior/functionalities, features of an application
- In order to fully test that all the requirements of an application are met, there must be *at least one test case for each requirement* (*two recommended*)

Test Case Format

- Test case ID
- Test case Description
- Expected Output
- Actual Output
- Pass/Fail
- Remarks
- Test step or order of execution number
- Related requirement(s)
- Depth
- Test category
- Author
- Check boxes for whether the test is automatable and has been automated.

Large Scale Engineering Projects

- Need a systematic approach:
  1. Choose a high level test plan
  2. Write detailed test cases
  3. Assign the test cases to testers, who manually follow the steps and record the results.
  4. Author a test report, detailing the findings of the testers.

- The report is used by managers to determine whether the software can be released

Test Automation

- **A process of writing a computer program to do testing** that would otherwise need to be done manually
- **The use of software to control the execution of tests**, the comparison of actual outcomes to predicted outcomes, the setting up of test preconditions, and other test control and test reporting functions
- Commonly, test automation involves automating a manual process already in place that uses a formalized testing process

Test Automation – Approaches

- **Graphical user interface testing**. A testing framework generates user interface events such as keystrokes and mouse clicks, and **observes the changes** that result in the user interface, to validate that the observable behavior of the program is correct
- **Code-driven testing**. The public (usually) interface to classes, modules, or libraries are tested with a variety of input arguments to validate that the results that are returned are correct

TA – What to test

- Testing tools can help automate tasks such as *product installation, test data creation, GUI interaction, problem detection, defect logging*, etc.
- Important points when thinking about TA:
  - Platform and OS independence
  - Data driven capability (Input Data, Output Data, Meta Data)
  - Customizable Reporting (DB Access, crystal reports)
  - Email Notifications
  - Easy debugging and logging
  - Version control friendly
  - Extensible & Customizable
  - Support distributed execution environment
  - Distributed application support

Test Automation – Example

QA Wizard - [Login (Functional_Tests)]

File Edit View Project Script Debug Database Tools Browser Window Help

A1 =URL

Project Explorer

Workspace | Reports

- WysiCorp_Software
  - Functional_Tests
    - Scripts
      - Smoke_Test
      - Login
      - Logout
      - Submit_Bug
      - Verify_Data
      - Initialize_Project_Variable
    - Database Environment
    - Project Variables
      - URL
      - USER
      - PASSWD
  - Regression_Tests
    - Scripts
      - Fill_DB
    - Database Environment
      - DBconnection
        - Defects_Recordset
    - Project Variables
  - Assemblies

Step #	Action Type	Object Type	Window
0	Main	main	
1	Open Browser		Window0
2	Navigate		Window0
3	Mouse Click	Text Field	Window0
4	Input	Text Field	Window0
5	Mouse Click	Password Field	Window0
6	Input	Password Field	Window0
7	Mouse Click	Submit Button	Window0
8	Navigate		Window0
9	Check Attributes	Bold Text	Window0

		A	B
	Iteration	Navigate Url	Input Text Field Value unam
1	main	http://www.seapine.com/wysicorp/login.	Tester

Previewer

siCorp What yo

Account Login

User Name:

Password:

Login Reset

Powered by Solo Submit

Test Automation with Scripting

- Manual Test Case Steps:
  - Launch Calculator
  - Press **2**
  - Press +
  - Press **3**
  - Press =
  - The screen should display **5**.
  - Close Calculator.

Example credit softwaretestinghelp.com

Test Automation with Scripting

```
//the example is written in MS Coded UI using c# language.
[TestMethod]
public void TestCalculator()
{

//launch the application
var app = ApplicationUnderTest.Launch("C:\\Windows\\System32\\calc.exe");

//do all the operations
Mouse.Click(button2);
Mouse.Click(buttonAdd);
Mouse.Click(button3);
Mouse.Click(buttonEqual);

//evaluate the results
Assert.AreEqual("5", txtResult.DisplayText,”Calculator is not showing 5);

//close the application
app.Close();
}
```

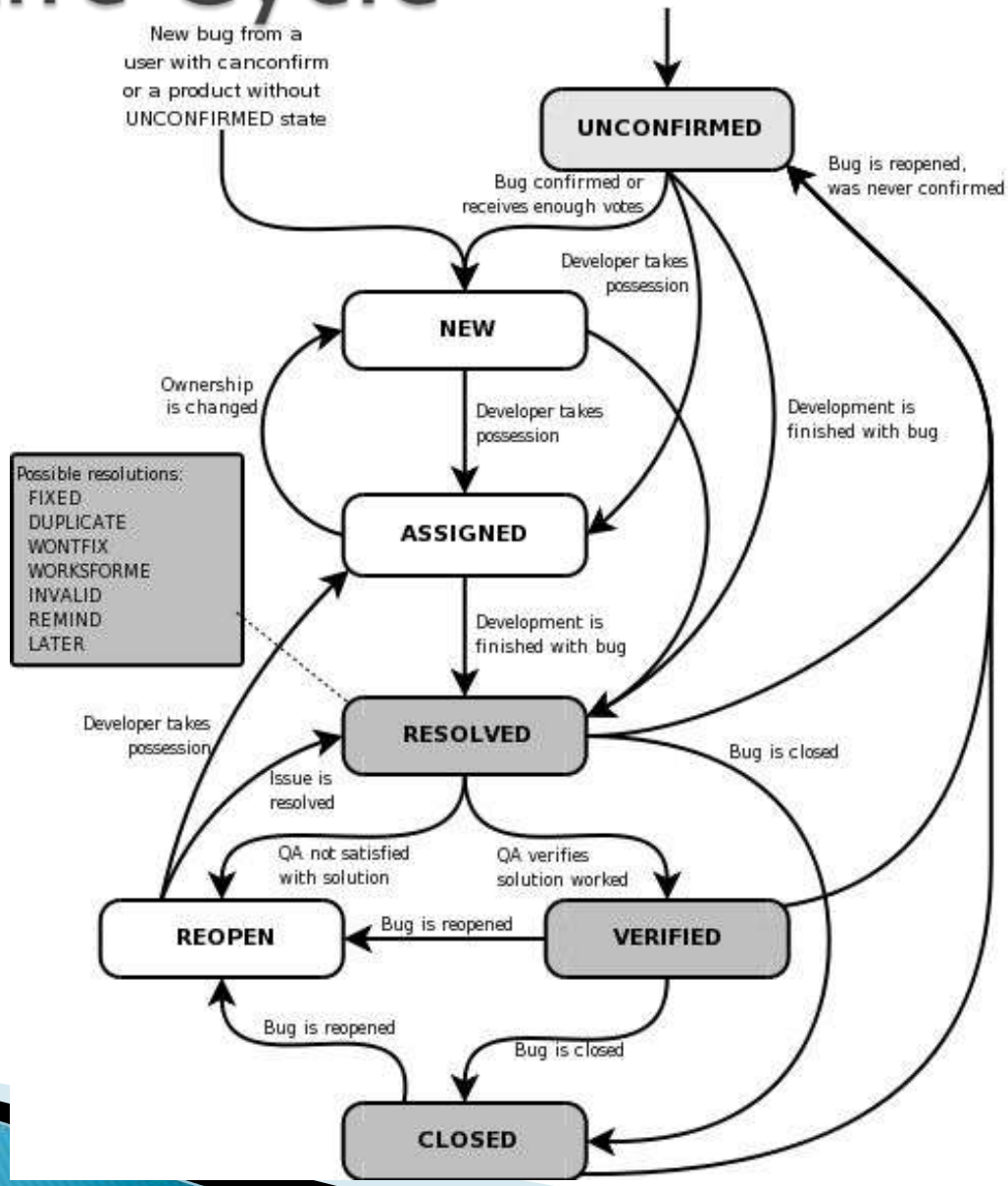
Software Bug

- **A software bug is an error, flaw, mistake, failure, or fault in a computer program**
- Most bugs arise from mistakes and errors made by people (in program or in its design), and a few are caused by compilers
- Reports detailing bugs in a program are commonly known as **bug reports**, **fault reports, problem reports, trouble reports, change requests**, and so forth.

Bugs Effects

- Bugs in the code controlling the Therac-25 radiation therapy machine were directly responsible for some patient deaths in the 1980s.
- Smart ship USS Yorktown was left dead in the water in 1997 for nearly 3 hours after a divide by zero error.
- Eve Online's deployment of the Trinity patch erased the boot.ini file from several thousand users' computers, rendering them unable to boot.
- Valve's Steam client for Linux could accidentally delete all the user's files in every directory on the computer.
- In 2002 were responsible for losses of $59 billion annually, or about 0.6 percent of the gross domestic product.

Bug Life Cycle



Bug Prevention

- Programming style
- Programming techniques
- Development methodologies
- Programming language support
- Code analysis
- Instrumentation

Coding Style – Motivation

- Coding conventions are important because:
  - 80% of the work on software is maintenance
  - Usually, a product is not maintained by the persons who created it
  - Code conventions improve code readability and allow a software developer to quickly understand new programs

Coding Style – General Reuquirements

- Use comments: what do functions do, what do variables represent, explain steps of algorithms, etc.
- Use suggestive names for variables and functions
- Develop modular code
- Use dual pairs: set/get, start/stop, add/remove, save/load

Correctness

- Correctness of an algorithm is asserted when it is said that **the algorithm is correct with respect to a specification**
- **Functional correctness** refers to the input-output behavior of the algorithm (i.e., for each input it produces the correct output)
- A distinction is made between **total correctness**, which additionally requires that the algorithm terminates, and **partial correctness**, which simply requires that *if* an answer is returned it will be correct.

Links

- Software Quality Assurance: http://satc.gsfc.nasa.gov/assure/agbsec3.txt
- Software Testing: http://en.wikipedia.org/wiki/Software_testing
- GUI Software Testing: http://en.wikipedia.org/wiki/GUI_software_testing
- Regression Testing: http://en.wikipedia.org/wiki/Regression_testing
- Junit Test Example: http://www.cs.unc.edu/~weiss/COMP401/s08-27-JUnitTestExample.doc
- https://www.test-institute.org/What_is_Software_Quality_Assurance.php

Coding Style – Links

- C++:
  - http://www.chris-lott.org/resources/cstyle/
  - http://geosoft.no/development/cppstyle.html
- Java:
  - http://java.sun.com/docs/codeconv/
  - http://geosoft.no/development/javastyle.html